

Add String with RE num1 = 456 p1 = 2 num2 = 77 p2 = 1

```
if (p1 < 0 || p2 < 0) { p1 = 2; p2 = 1; }
if (carry != 0)
    return String(c1, carry + '0');
return "0";
```

ans = '335'
Reverse the answer
533

```
int n1 = (p1 > 0 ? num1[p1] : '0') - '0'; n1 = 6;
int n2 = (p2 > 0 ? num2[p2] : '0') - '0'; n2 = 7;

int sum = n1 + n2 + carry; 6 + 7 + 0 = 13
int digit = (sum / 10); 3
carry = (sum / 10); 3
String ans = "";
ans.push_back(digit + '0');
```

p1 = 2 p2 = 0 (carry = 1)

```
ans += addRec(num1, p1 - 1, num2, p2 - 1, carry);
return ans;
ans = '3' + '35'
ans = '335'
```

```
int n1 = (p1 > 0 ? num1[p1] : '0') - '0'; n1 = 5;
int n2 = (p2 > 0 ? num2[p2] : '0') - '0'; n2 = 7;

int sum = n1 + n2 + carry; 5 + 7 + 1 = 13
int digit = (sum / 10); 3
carry = (sum / 10); 3
String ans = "";
ans.push_back(digit + '0');
```

```
ans += addRec(num1, p1 - 1, num2, p2 - 1, carry);
return ans;
ans = '8' + '35'
ans = '835'
```

p1 = 0 p2 = -1 (carry = 1)

```
if (p1 < 0 || p2 < 0) {
    if (carry != 0)
        return String(c1, carry + '0');
    return "0";
}
```

```
int n1 = (p1 > 0 ? num1[p1] : '0') - '0'; n1 = 4;
int n2 = (p2 > 0 ? num2[p2] : '0') - '0'; n2 = 0;

int sum = n1 + n2 + carry; 4 + 0 + 1 = 5
int digit = (sum / 10); 0
carry = (sum / 10); 0
String ans = "";
ans.push_back(digit + '0');
```

```
ans += addRec(num1, p1 - 1, num2, p2 - 1, carry);
return ans;
ans = '5'

p1 = -1 p2 = -1 (carry = 0)
if (p1 < 0 || p2 < 0) {
    if (carry != 0)
        return String(c1, carry + '0');
    return "0";
}
```

```
int n1 = (p1 > 0 ? num1[p1] : '0') - '0';
int n2 = (p2 > 0 ? num2[p2] : '0') - '0';

int sum = n1 + n2 + carry;
int digit = (sum / 10);
carry = (sum / 10);
String ans = "";
ans.push_back(digit + '0');
```

```
ans += addRec(num1, p1 - 1, num2, p2 - 1, carry);
return ans;
```